Shrikaran P

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Professional Summary

Computer Science Engineering student with strong expertise in Python, Deep Learning, PyTorch, and Machine Learning, backed by hands-on research in computer vision and Large Language Models. Experienced in developing CNN-based models, Retrieval-Augmented Generation (RAG) systems, and OCR-powered Local LLM pipelines through research projects and publications. Passionate about AI research, model optimization, and building production-ready intelligent systems with a strong foundation in deep learning, neural networks, and data-driven experimentation.

Technical Skills

- **Programming Languages**: Python, Java, C, C++
- **Core Concepts**: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)
- **AI/ML**: Machine Learning, Deep Learning, PyTorch, TensorFlow, Scikit-learn, CNN-LSTM, LLMs, RAG
- **Research Skills**: Literature Review, Model Training, Data Preprocessing, Hyperparameter Tuning, Model Evaluation, Experimentation
- **Databases**: SQL, MySQL, PostgreSQL
- **Backend & APIs**: FastAPI, REST APIs
- **Cloud & Tools**: AWS Cloud Fundamentals, Git, GitHub, Google Colab, Firebase
- **Web Technologies**: HTML, CSS
- **Soft Skills**: Problem Solving, Attention to Detail, Team Collaboration, Leadership, Communication, Quick Learner, Time Management

Educational Qualifications

B.Tech in Computer Science Engineering(6th semester) May 2023 - May 2027
SRM Institute of Science and Technology - Tiruchchirappalli
CGPA - 9.3/10

Achievements

- **3rd Prize Winner (Cash Award)** – Ideathon, SRM Institute of Science and Technology, for developing a Centralized Dam Management System.
- **Participant** – 24-Hour Hackathon, Dhanalakshmi Srinivasan University, where I developed a Farmer Market Price Prediction System as part of a team.
- **Foundations of AI/ML – GeeksforGeeks -** Gained hands-on experience in data preprocessing, model building, and evaluation techniques
- **Java Full Stack Development – EduSkil**l - Built web-based applications with structured data handling and backend integration
- **AWS Academy Cloud Foundations – AWS**
  Learned fundamentals of cloud computing and data storage for scalable data processing

Research Internship(NIT Trichy)

- Motor Imagery EEG Classification (BCI) — Worked with BCI Competition IV Dataset 2a and BCI Competition IV Dataset 2b EEG .gdf recordings for motor imagery analysis.
- Preprocessed raw EEG signals, selected motor cortex channels, and generated STFT-based time–frequency spectrogram representations.
- Developed CNN-based models for left/right hand motor imagery classification and subject-wise performance evaluation.
- Trained and evaluated deep learning models using PyTorch, improving classification performance through hyperparameter tuning and experimentation.

Projects

A Secure On-Premises ETL Pipeline Using OCR and Local LLMs

- Designed a secure ETL pipeline to extract text from scanned documents using OCR and transform unstructured data into structured formats with Local LLMs.
- Automated document processing, data validation, and information extraction while ensuring data privacy and improving processing efficiency.

Contrastive Knowledge Graph-RAG for Concept Misconceptions

- Engineered a Knowledge Graph-based Retrieval-Augmented Generation (KG-RAG) system using contrastive learning to detect and explain concept misconceptions.
- Integrated embeddings, vector retrieval, and Large Language Models to improve retrieval accuracy, contextual understanding, and explainable AI responses.

Weather Data Pipeline API

- Built a FastAPI-based REST API to ingest real-time weather data from the OpenWeather API, process it, and store it in PostgreSQL for efficient data management.
- Implemented secure API endpoints with filtering, validation, and historical weather retrieval, enabling reliable data access for analytics and reporting.

Intrusion Detection System using CICIDS2017

- Designed a deep learning–based intrusion detection model to classify malicious and benign network traffic using the CICIDS2017 dataset.
- Compared model performance against traditional rule-based methods using accuracy, precision, recall, and F1-score metrics.

Farmer Market Price Prediction System

- Created a machine learning model to predict crop market prices using historical mandi data, seasonal trends, weather conditions, and demand–supply analysis.

- Provided predictive analytics and market insights to support informed selling decisions and improve farmers' profitability.

Research & Publications

- A Secure On-Premises ETL Pipeline Using OCR and Local LLMs - ICCMC 2025, Tiruppur
- Contrastive Knowledge Graph-RAG for Concept Misconceptions-RECCAP 2026, IIT Palakkad